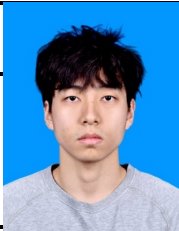


PERSONAL INFORMATION

Name: Zhenhao Tian **Ethnicity:** Han **Email:** tian-zh24@mails.tsinghua.edu.cn
Phone: 13509612319 **Birth:** Mar 2002



EDUCATION

B.Eng. in Cyberspace Security, Xu Teli School, **Beijing Institute of Technology** **Sep 2020 - Jun 2024**

- Ranked in the top **4.8%** of the school; GPA: **3.8/4.0**; CET-4: 601; CET-6: 549.
- **Core courses:** Mathematical Analysis (95), Data Structures and Algorithm Design (95), Digital Logic (99), Probability and Mathematical Statistics (100), Cyberspace Security Course Design (99), Mathematical Foundations of Cyberspace Security (96).
- **Honors:** Tsinghua University First-Class Comprehensive Excellence Scholarship; BIT Academic Scholarship; BIT Outstanding Student Award.

M.S. in Cyberspace Security, Institute for Network Sciences and Cyberspace, **Tsinghua University** (recommended admission; expected 2027) **Sep 2024 - Jun 2027**

- **Research interests:** **LLM security** and security applications, **firmware security**, and **vulnerability discovery**.

RESEARCH EXPERIENCE

LLM-Based Internet-Connected Device Asset Identification (CCF-A Journal TIFS, First Author) DOI: 10.1109/TIFS.2025.3587211

- Designed and developed a **dual-round LLM role mechanism** for asset discovery and expansion, enabling structured data processing and output through LLM-based extraction and decision-making.
- Worked on device entity recognition using **corpus-based methods, regular-expression matching, and rule-based recognition from training data**; extracted device fingerprint features from collected data using NLP techniques and produced device asset information.

Static Vulnerability Discovery for IoT Devices Based on LLM-IR (Submitted to CCS, First Author) **Nov 2025 - Present**

- Developed automated **taint-label identification** and **rule-constrained false-positive reduction** modules by combining CodeQL query paradigms with LLM-based code semantic understanding; identified sources in device backend services such as dhcpd and upnp.
- Built a path-level false-positive reduction algorithm that migrates manual auditing practices into automated module auditing, including reachability checks based on **function pairs/backward slicing**.
- Experimentally discovered **130+ vulnerabilities** across mainstream IoT vendors' devices, with **38+ CVE IDs** assigned to date.

Router Vulnerability Threat Knowledge Graph **Sep 2023 - Dec 2024**

- Collected and studied vulnerability assessment and scanning scripts including OVAL and NASL, and performed **data correlation and cleaning**.
- Ported sh/rb/yaml scripts into a **unified code framework** and built local vulnerability environments for **PoC effectiveness verification**.
- Designed and developed an automated router vulnerability scanning and verification agent, using **message queues** to support concurrent multi-service vulnerability probing.

Open-Source Package Poisoning Detection **Apr 2023 - Jun 2023**

- Conducted source-code security research on Python and JavaScript package ecosystems; performed two months of **static code auditing** and classified malicious behavior from package source code.
- Studied package dependency relationships and wrote scripts based on dependency call graphs to analyze **code execution sequences**, including snippets, hierarchy, and execution types.

COMPETITIONS AND AWARDS

Champion, 2025 Qiangwang Cup Vulnerability Special Track **Oct 2025 - Nov 2025**

- Migrated the IRIS specification paradigm to other languages to automatically generate CodeQL specification queries; completed offline blind testing and defense.

Second Prize, 2025 DataCon AI Security Track **Nov 2025 - Dec 2025**

- Designed semantic-axis methods to extract RAG knowledge chunks and used coloring algorithms to derive standardized knowledge rules.

INTERNSHIP EXPERIENCE

Tencent Xuanwu Lab: Cybersecurity & Multi-Agent Cyber Intern	Apr 2026 - Jul 2026
Huawei Singularity Lab: Cybersecurity & LLM Application Development Intern	Jul 2025 - Sep 2025
ByteDance Douyin Red Team: Cybersecurity & LLM Application Development Intern	Mar 2025 - Jun 2025
Security Institute, CAICT: Software Security Research / Cybersecurity Intern	Mar 2023 - May 2023
Zhongguancun Laboratory: Cybersecurity Intern	Mar 2024 - Jun 2024